15

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CSE 340 Computer Architecture

Spring 2023

Quiz 1

Marks: 15

Time: 30 mins

ANSWER ALL THE QUESTIONS

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Section 03

C01

1. Suppose you are asked to choose the best performing CPU. [5]

Specification of Machine A -

Instruction count = 5 x 10^9, Clock rate = 300MHz, CPI= 3.2

Specification of Machine B -

Instruction count = 2 x 10^9, Clock rate = 600MHz, CPI= 4.8

a. State which one is faster and by what factor.

b. State how CPU time is affected by changes in CPI and Clock Period.

(a) $CPU_A = \frac{5\times10^9 \times 3.2}{300\times10^6} = 53.33s$ $CPU_B = \frac{2\times10^9\times4.8}{600\times10^6} = 16s$

$\frac{Per_B}{Per_A} = \frac{CPU_A}{CPU_B} = \frac{53.33}{16} = 3.33$

So, CPU_B is faster 3.33 times as it's CPU time ~~is~~ less.

b. CPU time = no of instruction × CPI × clock Period.

So, increase in CPI and clock period will ~~affect~~ increase affecting to CPU time of CPU time. So, it is proportionally

C01

2. Consider the following classes of tasks for a particular CPU performing a task.

	IC (10^9)	CPI
Task A	3	2.6
Task B	1	1.7

Task C	5	2.3
Task D	1	3.1

a. Calculate the average CPI [3]

b. Suppose it has a clock rate of 1.8MHz, then what is the CPU time. [3]

c. If the program runs on a reference PC with an execution time of 120 nanoseconds, calculate the SPECRatio. [2]

d. State two differences between Von Neumann Architecture and Harvard Architecture. [2]

a. $$\text{avg CPI} = \frac{(3\times 2.6)+(1\times 1.7)+(5\times 2.3)+(1\times 3.1)}{3+1+5+1}$$

$$= 2.41$$

b. $$\text{CPU time} = \frac{(3\times 2.6 \times 10^9)+(1\times 1.7\times 10^9)+(5\times 2.3 \times 10^9)+(1\times 3.1 \times 10^9)}{1.8\times 10^6}$$

$$= 13,388.89\ s$$

(c) $$\text{SPECRatio} = \frac{120\times 10^{-9}s}{13388.89s} = 8.963\times 10^{-12}$$

d.

Von Neumann Architecture	Harvard Architecture
(i) ~~Data memory~~ Shared memory by data and instruction	(i) Data memory and instruction memory is different
(ii) Can not access both at a same time, so sequential access to memory.	(ii) Can access both at a same time, so parallel access to memory.